

Dexter Fernandes

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Senior Data Scientist

Computer Vision & Machine Learning Engineer with 3+ years delivering perception, predictive modelling and analytics pipelines in safety-critical and insurance settings. Built production-grade systems for low-latency inference, scalable data processing and end-to-end model development. Developed an automotive claims pipeline with damage detection, severity classification and insurer-ready interpretability. Proven track record of leading engineering teams, mentoring juniors, and integrating AI modules into Dockerised workflows. Eager to contribute to Verisk's mission by developing reliable, interpretable AI systems that enhance claims workflows and vehicle-assessment accuracy.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Built predictive models for classifying 24/7 oil-rig operations using time-series and CV features, achieving 23% higher F1 and supporting automated risk triage.
- Developed real-time perception modules for oil-rig safety systems, deploying CNN-based vision pipelines with YOLOv5 on edge devices to improve detection accuracy by 15%.
- Implemented automated data ingestion and cleaning workflows for large, multimodal sensor datasets using MLFlow, DVC, and cloud storage, streamlining experimentation and enabling faster model iteration.
- Integrated AI modules with production systems in collaboration with other teams to deliver fault-tolerant edge deployments.
- Designed automated data collection, augmentation, and labeling tools to scale large image datasets, increasing dataset diversity by 40% and cutting manual labeling time by 60%.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Built synthetic image generation and augmentation procedures to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Deployed quantised OpenVINO vision models over Microsoft Azure to rugged Intel VPU edge devices for on-site inference in harsh industrial environments.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50%.
- Implemented rigorous testing and validation frameworks simulating challenging lighting, occlusion, and motion conditions, improving CNN stability and reducing variance by 25%.
- Mentored engineers on optimisation and deployment practices, raising code quality and delivery consistency.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

PROJECTS

End-to-End Damage Severity & Claim Assessment

- Developed an end-to-end automotive claims assessment pipeline combining computer vision-based damage detection with a trained severity-classification model, achieving accurate triage for real-world vehicle damage scenarios.
- Built a lightweight repair-insight engine that transforms model outputs into insurer-friendly explanations, using structured reasoning + LLM-generated narratives to deliver clear, interpretable claim summaries.
- Designed the entire workflow (data preprocessing, model training, evaluation, and a user-facing inference module) demonstrating practical understanding of insurance claims processes.

SKILLS

- **Perception & SLAM:** Data Augmentation, NumPy, OpenCV, SAM, Scikit-Image, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, Docker, DVC (Data Version Control), FastAPI, Git, Linux, Microsoft Azure, MLFlow
- **Programming & Development:** Bash, C++, Pandas, PostgreSQL, Python
- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, ONNX, OpenVINO, Optuna, PyTorch, Quantisation-Aware Training, TensorFlow, TensorRT